

Lesson 41 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

Provenance: All 25 items are original exam-pattern questions (*[Original]* plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

- Q1. B.** TCP stands for Transmission Control Protocol. A, C, and D are not the correct expansion.
- Q2. A.** DNS stands for Domain Name System. B, C, and D are not the correct full forms.
- Q3. C.** TCP is reliable, ordered, and connection-oriented. A describes UDP, not TCP; B describes TCP, not UDP; D wrongly promises UDP timing guarantees.
- Q4. B.** TCP guarantees complete, ordered delivery through sequencing and retransmission; it does not promise speed (A), encryption (C), or name resolution (D).
- Q5. C.** UDP gives no ordering guarantee, so this statement is NOT true. Connectionless operation (A), best-effort delivery (B), and use for streaming and DNS (D) are all true of UDP.
- Q6. B.** IPv4 uses 32-bit addresses and IPv6 uses 128-bit addresses (TRM-10). A reverses them; C and D use the 256-bit distractor (TRAP-18).
- Q7. C.** IPv6 is 128-bit with a fixed 40-byte header (PYQ-18). 256-bit (A) is the known trap; four decimal octets (B) describe IPv4; a 20–60 byte header (D) describes IPv4.
- Q8. B.** ICMP carries control and diagnostic messages (errors, reachability) only. It is not a file-transfer (A), mail-sending (C), or encryption (D) protocol.
- Q9. A.** Ping and traceroute are the two ICMP diagnostic tools. Mail merge and autosum (B, D) are Office features; FTP and Telnet (C) are separate application protocols.
- Q10. C.** DNS translates a domain name into its IP address. DHCP (A) assigns addresses; ICMP (B) reports errors; SMTP (D) sends mail.
- Q11. B.** Automatically handing out the address, mask, gateway, and DNS server is DHCP. DNS (A) resolves names; ICMP (C) diagnoses; FTP (D) moves files.
- Q12. C.** HTTP is port 80 and plain text. A gives HTTP the HTTPS port and property; B reverses the pair; D gives HTTPS the SMTP port and role.
- Q13. A.** SMTP (Simple Mail Transfer Protocol) sends outgoing mail. Downloading (B) is POP3; syncing on the server (C) is IMAP; name translation (D) is DNS.
- Q14. B.** POP3 downloads received mail to the local machine. A gives SMTP the IMAP role; C gives IMAP the SMTP role; D gives SMTP the DNS role.

- Q15. B.** Statements I and II are correct: TCP is connection-oriented and ordered, and IPv6 is 128-bit. III is false because ICMP transfers no application data and uses no ports (TRAP-19). A omits II; C includes false III and omits I; D includes false III.
- Q16. C.** HTTP, FTP, SMTP, and DNS are Application-layer (layer 7) protocols. TCP and UDP belong to the Transport layer, not the Network layer (A); IP and ICMP belong to the Network layer, not Transport (B); the Physical layer (D) carries bits, not IP or HTTP.
- Q17. B.** The OSI order is Physical, Data Link, Network, Transport, Session, Presentation, Application. A is the exact reverse; C and D scramble the middle layers.
- Q18. B.** SMTP is port 25 and DNS is port 53. A swaps FTP (21) and Telnet (23); C swaps HTTP (80) and HTTPS (443); D swaps POP3 (110) and IMAP (143).
- Q19. D.** ICMP uses no ports and transfers no web pages, so this is INCORRECT. FTP on 21 (A), Telnet on 23 (B), and DHCP address assignment (C) are all correct.
- Q20. A.** A successful ping proves the Network-layer path works (ICMP control traffic), so the fault lies above it: DNS, proxy, or application settings. B and C wrongly treat ICMP as page delivery; D names a layer that does not exist.
- Q21. A.** TCP/IP carries Internet traffic. B, C, and D name unrelated materials or layouts.
- Q22. A.** IPs label network devices. B, C, and D are non-network identifiers.
- Q23. A.** IPv4 uses 32 bits. B is too short; C is IPv6's length; D is not an IP size.
- Q24. A, B, C.** TCP/IP, HTTP, and FTP are protocols. A stapler pin (D) is not.
- Q25. A, B, C.** The TCP, IP, and FTP roles are correct. D is false: IPs identify devices, not pages.
- Q26. B.** The S in HTTPS means Secure. The other options are not expansions of HTTPS.
- Q27. A.** W3C is the World Wide Web Consortium. The other organizations have different mandates.
- Q28. A.** TCP provides reliable, ordered transport. UDP is connectionless without that delivery guarantee; ARP and ICMP serve other network functions.
- Q29. A.** DNS resolves names. SMTP sends mail, FTP transfers files, and POP3 retrieves email.
- Q30. A, B, C.** HTTP, SMTP, and FTP are application-layer protocols. Ethernet framing belongs to the data-link layer.